

Logic Gates and Universal Gates

Digital System Design - Unit 1: Logic Gates and Boolean Algebra - Session 3

1. Digital signals

A digital circuit works with two voltage levels: LOW (logic 0, about 0 V) and HIGH (logic 1, about 5 V in TTL or 3.3 V in modern CMOS). A **logic gate** is an electronic circuit whose output is a Boolean function of its inputs. Every digital system, from a calculator to a processor, is built from a handful of gate types.

2. Basic gates

Gate	Boolean expression	Output is 1 when
AND	$Y = A \cdot B$	all inputs are 1
OR	$Y = A + B$	at least one input is 1
NOT	$Y = A'$	the input is 0
XOR	$Y = A \oplus B = A'B + AB'$	the inputs differ

3. Truth table of two-input gates

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

NAND and NOR are the complements of AND and OR: $Y = (A \cdot B)'$ and $Y = (A + B)'$.

4. Universal gates

NAND and NOR are called **universal gates** because any Boolean function can be built using only one of them. With NAND gates alone:

- NOT: tie both inputs together, $(A \cdot A)' = A'$.
- AND: a NAND followed by a NAND-inverter, $((A \cdot B)')' = A \cdot B$.
- OR: invert each input first, $(A' \cdot B')' = A + B$ (by De Morgan's theorem).

5. De Morgan's theorems

$$(A \cdot B)' = A' + B' \quad (A + B)' = A' \cdot B'$$

The complement of a product is the sum of the complements, and the complement of a sum is the product of the complements.

6. Worked example: XOR from four NAND gates

1. $N1 = (A \cdot B)'$
2. $N2 = (A \cdot N1)'$
3. $N3 = (B \cdot N1)'$
4. $Y = (N2 \cdot N3)'$

Check $A = 1, B = 0$: $N1 = 1, N2 = (1 \cdot 1)' = 0, N3 = (0 \cdot 1)' = 1, Y = (0 \cdot 1)' = 1$, which matches $1 \oplus 0 = 1$.

7. Application: half adder

A half adder adds two bits: $\text{Sum} = A \oplus B$ and $\text{Carry} = A \cdot B$. Adding $1 + 1$ gives Sum 0, Carry 1, that is binary 10.

8. Common misconceptions

'XOR is just OR': the two differ when both inputs are 1 (OR gives 1, XOR gives 0). 'NAND is the opposite of OR': NAND is the complement of AND; by De Morgan, $(A \cdot B)' = A' + B'$.

9. Practice

1. Write the truth table of a three-input AND gate.
2. Implement $Y = A + B$ using only NOR gates.